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DIAGNOSTIC METHODS: OBSERVATIONAL STUDY

Abdominal and pelvic floor electromyographic analysis during abdominal hypopressive gymnastics

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ABSTRACT

Introduction: Abdominal hypopressive gymnastics appeared as an alternative to traditional abdominal exercises to promote abdominal muscles strength without overloading the pelvic floor muscles (PFM). To determine the activation level of abdominal muscles and PFM and the posture influence in the level of activation in these muscles during abdominal hypopressive gymnastics, we used surface electromyography in young and healthy multipara women.
Methods: This is an observational study with eutrophic nulliparous women aged between 18 and 35 years, with abdominal skinfold less than or equal to 3 cm and active or irregularly active physical activity. Surface electromyography was used for rectus abdominis, external oblique, transversus abdominal/internal oblique (TrA/IO) and PFM assessment in the supine, quadruped and orthostatic (upright standing) positions during abdominal hypopressive gymnastics using normalized electromyographic (%EMG) data. We also analyzed the difference in activation between each muscle and between muscles and positions.
Results: Thirty women were evaluated and the mean age was 25.77 years (SD 3.29). The group formed by the TrA/IO muscles and the PFM showed higher %EMG in all the positions assessed, followed by the external oblique and rectus abdominis muscles. A comparison of %EMG of each muscle between the different positions showed differences only in rectus abdominis between the supine and quadruped ($p = 0.001$) and supine and orthostatic positions ($p = 0.004$), and in TrA/IO between the supine and orthostatic ($p = 0.023$) and orthostatic and quadruped positions ($p = 0.019$).
Conclusions: The results suggest that abdominal hypopressive gymnastics can activate the abdominal muscles and PFM and the position do not have influence on electromyographic activation level of the PFM and external oblique.

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1. Introduction

Abdominal hypopressive gymnastics (AHG), developed in the 1980s, is an alternative technique for strengthening abdominal (AM) and pelvic floor muscles (PFM) in the postpartum period (Caufriez, 1989, 1988; Caufriez et al., 2007, 2006). This gymnastics promotes reflex activation in these muscle groups in response to negative intra-abdominal pressure (IAP) (Caufriez, 1989, 1988). Negative IAP results from contraction of the accessory muscles of respiration during expiratory apnea, causing an opening of the rib cage and the diaphragmatic cupula to move forward (Caufriez et al., 2011, 2007; Rial and Villanueva, 2012). This technique is performed in different static or dynamic positions, since each has a degree of difficulty and specific activation of muscle groups (Bernardes et al., 2012; Caufriez et al., 2011, 2007, 2006).

Since its inception, it has been used not only in the postpartum but also in the conservative treatment of urinary and fecal incontinence and genital prolapse (Caufriez, 1995). Moreover, it has been used as a postural technique in the treatment of back pain, as well as in gym activities, using a set of rhythmic and sequential exercises (Rial and Villanueva, 2012; Torres and Salido, 2009).

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Studies have assessed the effect of AHG on AM and PFM activation, or its clinical repercussions (Bernardes et al., 2012; Caufriez et al., 2011, 2007, 2006; Resende et al., 2012; Stüpp et al., 2011; Torres and Salido, 2009). Only one of these investigated the electromyographic activity of these muscles (Stüpp et al., 2011), analyzing, only in the supine position, if this type of exercise was able to promote PFM activation proportional to that observed in free active exercise of these muscles. Thus, although this technique has existed for 25 years, there is a gap in the literature regarding its effect on muscle activation.

Regarding to synergistic activation of PFM and AM, surface electromyography has been widely used as a useful and reliable tool, capable of quantifying muscle recruitment level, providing more accurate data on the association between these muscles during these exercises (Auchincloss and McLean, 2012; Chanthapetch et al., 2009; Keshwani and McLean, 2013; Stéphanie J Madill and McLean, 2008; Marshall and Murphy, 2003; Neumann and Gill, 2002; Sapsford et al., 2001; Sapsford and Hodges, 2001; Taillon-Hobson et al., 2011; Vera-Garcia et al., 2011).

Therefore, the present study aimed at investigating whether abdominal hypopressive gymnastics can promote abdominal and pelvic floor muscle activation, in addition to identifying the relative activation level of these muscles and the association between body position and muscle response during AHG.

2. Materials and methods

2.1. Subjects

This is an observational study with eutrophic nulliparous women aged between 18 and 35 years (Body Mass Index -BMI- between 18.50 and 24.99 kg/m^2), with abdominal skinfold less than or equal to 3 cm (measured 4 cm to the right of the umbilical scar using a Sanny™ adipometer, with pressure of 10 kg mm^2) and active or irregularly active physical activity, assessed using the short form International Physical Activity Questionnaire (IPAQ) (Matsudo et al., 2001). The IPAQ was used to decrease the influence of the physical activity level on the electromyographic signal amplitude. The level of physical activity classification followed the criteria available in Matsudo et al. (2001) (Matsudo et al., 2001). The following exclusion criteria were adopted: history of abdominal or pelvic surgery, pregnant women or suspected pregnancy, hypertension, diabetes, history of smoking, neurological, respiratory or cardiac disease, pelvic floor or menstrual dysfunctions. Participants were recruited after the study was approved by the Human Research Ethics Committee of the Federal University of Pernambuco, under CAAE protocol 0136.0.172.000–11, and those who agreed to participate gave their informed consent.

2.2. Abdominal hypopressive gymnastics

All the volunteers received previously abdominal hypopressive gymnastics training, a week before the data collection. The subjects were instructed to start the exercise with costal breathing and raising the lower ribs, followed by total expiration. They were then asked to perform expiratory apnea and, with the glottis closed, expanding and raising their rib cage by contracting accessory inspiratory muscles (serratus anterior, intercostals, scalene and sternocleidomastoid) (Caufriez et al., 2007, 2006; Rial and Villanueva, 2012). The following positions were analyzed: orthostatic (upright standing), quadruped and supine (see Fig. 1), as described by Caufriez et al. (2006) (Caufriez et al., 2006).

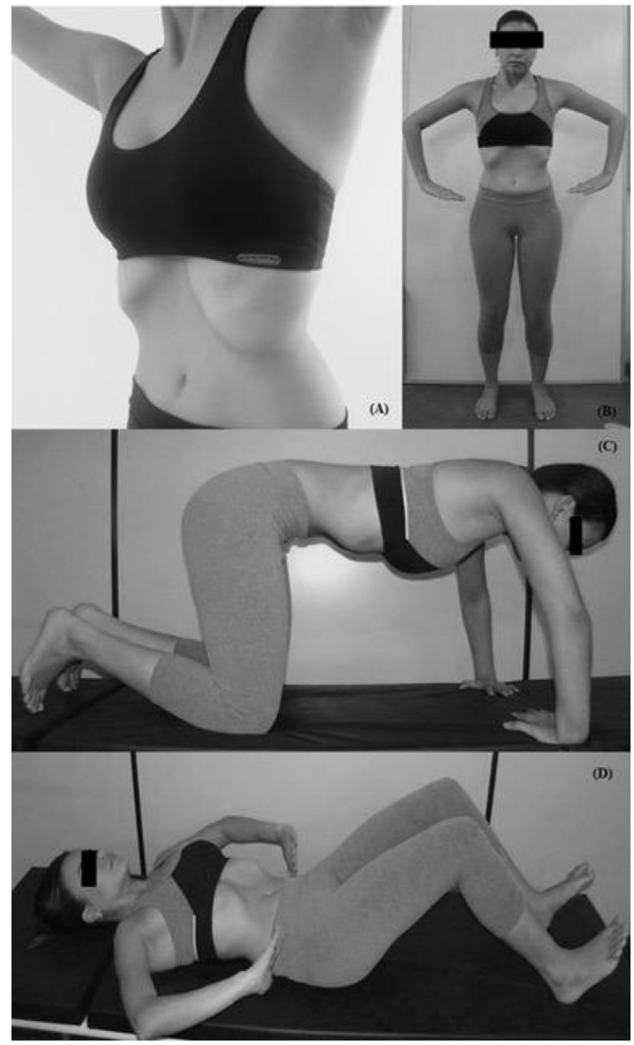

Fig. 1. - Expansion of the chest by opening the rib while performing (A) Abdominal hypopressive gymnastics, (B) Orthostatic, (C) Quadruped, (D) Supine.

2.3. Instrumentation

To acquire the biological signal, the Miotec™ Miotol 400 (four channel) surface electromyograph (Porto Alegre, Rio Grande do Sul, Brazil) was used, with analog/digital (A/D) convertor (14-bit resolution), internal gain of 1000 times, sampling frequency 2000 Hz, common-mode rejection ratio (CMRR) 110 dB, input impedance 10^{10} Ω//2 pF, and Butterworth fourth order analog filter. 20 Hz high-pass, 500 Hz low-pass and 60 Hz notch filters were applied.

Electromyographic signals were acquired through surface electrodes. The bipolar technique was used to collect abdominal muscle signals, where two low-impedance disposable AG/AgCI double junction surface disk electrodes (Meditrace™ disposable pediatric cardiac electrodes), made from hypoallergenic highly adhesive medical tape and adhesive solid gel (Hydrogel), were placed 25 mm apart on each muscle (Ng et al., 1998). Electromyographic assessment of the pelvic floor was conducted using an intravaginal probe (Miotec™, Porto Alegre, Rio Grande do Sul, Brazil). A reference electrode was placed on the right anterior superior iliac spine in order to eliminate external interference (Stéphanie J Madill and McLean, 2008).

2.4. Data acquisition

The following muscles were assessed: the rectus abdominis (RA), the external oblique (EO), the group formed by the transverse abdominal and internal oblique (TrA/IO) muscles, in addition to pelvic floor muscles (PFM). The TrA/IO muscle group was used because the IO cannot be differentiated from the TrA due to anatomic fusion in the region used for signal acquisition (Marshall and Murphy, 2003). The electrodes were positioned along the action line of the right RA muscle fibers (2 cm lateral and caudal to the navel), right EO (on the eighth rib, angled diagonally in the direction of the muscle fibers), and the right TrA/IO (2 cm proximal to the midpoint on the line between the anterior superior iliac spine and the pubic symphysis) (Neumann and Gill, 2002). The skin was cleaned with alcohol before the electrodes were placed on each anatomic site. For PFM assessment, the subjects were instructed to insert the intravaginal probe, which was done in a private room. When there was a doubt regarding positioning of the probe, a researcher performed an inspection. A reference electrode was used on the anterior superior iliac spine (see Fig. 2).

To normalize the signals, the maximum voluntary contraction (MVC) was recorded for each muscle, using the following standardized tests as a reference measure: Maximum voluntary isometric trunk flexion for the RA, contralateral trunk rotation for the EO (Hodges et al., 1999) and the abdominal retraction maneuver for the TrA, where the subject performs a movement that brings the navel toward the spine. For PFM normalization, the participants were instructed to contract them, following the instruction: "Pull up and in, and squeeze around the probe" (Stéphanie J Madill and McLean, 2008). All the tests were conducted in the supine decubitus position.

After all the muscles were assessed during MVC, we then register the electromyographic signal during AHG (MVC AHG) in the supine, quadruped and orthostatic position. The order of positions was randomly defined using Microsoft Excel 2010™ software to minimize the effect of muscle fatigue on electromyographic signal amplitude. Three 15-second recordings were made in each position, maintaining expiratory apnea with the lower ribs raised, for all activities analyzed. Subjects rested for at least 60 s between each activity.

2.5. Data processing and analysis

2.5.1. Signal processing

The electromyographic data obtained were analyzed by Miograph™ software (Miotec™, Porto Alegre, Rio Grande do Sul, Brazil). Initially the electromyographic record (raw) was inspected (see Fig. 3) to select the sections that showed high electrical activity without the presence of R waves of electrocardiogram (ECG). Afterwards sections of 500 ms (ms) was selected, filtered (20 Hz high pass, 500 Hz low pass, 60 Hz notch), converted to root mean square (RMS) and smoothed in mobile rectangular window of 50 ms. The EMG corresponding to the peak EMG in microvolts (μV) during the maximum voluntary contractions (MVC) and AHG (MVC AHG) was used to perform the calculations and comparisons.

To normalize the electromyographic signal, the MVC AHG of each muscle and in each position was divided by the reference MVC, in order to express the activation level of muscles during AHG as a percentage of the greater activation of each muscle obtained through standardized tests.

2.6. Data analysis

The Kolmogorov-Smirnov was applied to determine if the variables exhibited normal distribution. Due to variable non-normal distribution it was used a non-parametric test. The Friedman test was used to compare the muscles electromyography activity (dependent variables) across different positions (independent variables) or compare positions for each muscle separately, and the post hoc Wilcoxon test for related samples to identify specific differences. The p-value was adjusted for the number of pairwise comparisons when using post-hoc Wilcoxon test. A 5% significance level was set in all the tests. Statistical analyses were carried out using Statistical Package for Social Sciences (SPSS) software, version 18.

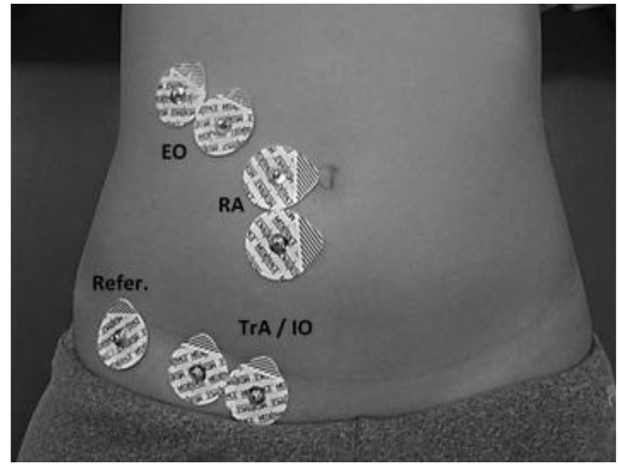

Fig. 2. – Electrodes positioning on the rectus abdominis (RA), external oblique (EO), transversus abdominis/internal oblique (TrA/IO) and reference electrode (Ref.).

3. Results

Thirty individuals, mean age 25.77 years (SD 3.29), body mass 56.05 kg (SD 6.72), height 1.62 m (SD 0.07) and body mass index 21.11 kg/m^2 (SD 1.64), took part in the study (see Fig. 4). According to IPAQ, 63.4% showed irregularly active physical activity and no subjects had history of backache.

Table 1 shows the maximum median electromyographic activity levels obtained during standardized tests, during abdominal hypopressive gymnastics, median normalized electromyographic activity and percent variation in relation to pre-exercise activity of each muscle in each position.

The group formed by the transverse abdominal and internal oblique (TrA/IO) muscles, and pelvic floor muscles (PFM) showed higher activation percentages (%EMG) in all the positions assessed, followed by the external oblique (EO) and rectus abdominis (RA) muscles (see Table 2).

A comparison of %EMG of each muscle between the different positions showed differences only in RA between the supine and quadruped ($p = 0.001$) and supine and orthostatic positions ($p = 0.004$), and in TrA/IO between the supine and orthostatic ($p = 0.023$) and orthostatic and quadruped positions ($p = 0.019$) (see Fig. 5).

4. Discussion

The present study, which investigated the behavior of abdominal and pelvic floor muscles during abdominal hypopressive gymnastics, is the first to assess the effect of body position on the electromyographic activity of these muscles during AHG.

Abdominal and pelvic floor muscle recruitment showed small variations between the positions analyzed. The TrA/IO and PFM exhibited higher relative activation percentage, followed by the EO

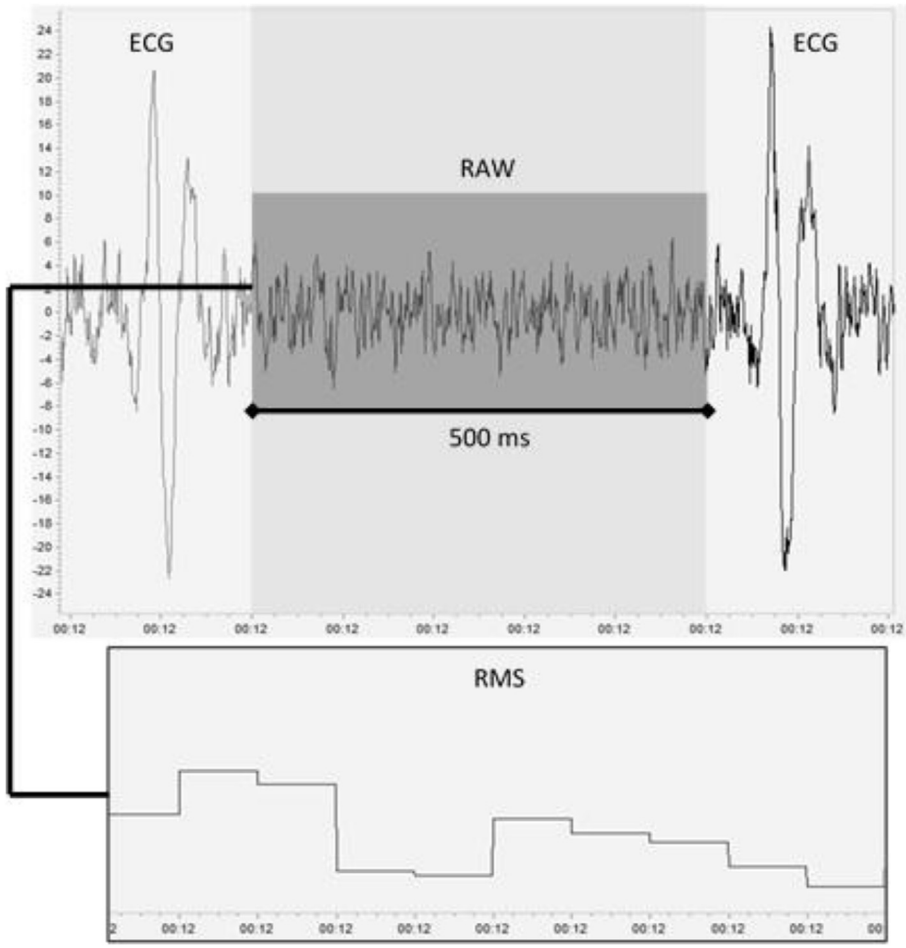

Fig. 3. - Signal processing: Root mean square (RMS) of RAW in a section of 500 milliseconds (ms) among the electrocardiogram signals (ECG).

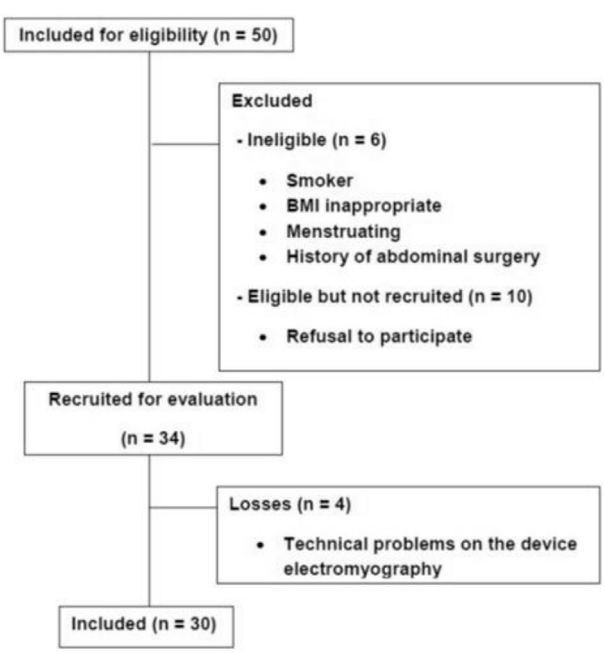

Fig. 4. - Flowchart of uptake and monitoring of participants.

and RA, indicating greater synergism between the TrA/IO and PFM during AHG when compared with EO and RA. Literature reports regarding the activation level of these muscles during maximum voluntary contraction of the PFM revealed similar results (Stéphanie J. Madill and McLean, 2008). In this study, which assessed adult women with no history of urinary stress incontinence, the authors suggest that the TrA and IO are more closely related to PFM activity than the EO and RA. This close relation was also observed during the execution of respiratory maneuvers such as coughing and forced expiration by adult nulliparous women (Neumann and Gill, 2002). According to Junginger et al. (2010) (Junginger et al., 2010), the co-contraction existing between the TrA and PFM can be observed in submaximal efforts.

The low percentage of RA activation, ranging from 5.44% to 9.77% between positions, may indicate a weak relation between AHG and activation of this muscle, in addition to the low synergism between the RA and the other muscles assessed. This lower activation may be related to the differences in anatomic fixation of these muscles, sharing minimum attachments at their origins and insertions (Chanthapetch et al., 2009).

In this study, only the normalized electromyographic activity of the RA (supine vs. quadruped and supine vs. orthostatic) and the TrA/IO (supine vs. orthostatic and orthostatic vs. quadruped) exhibited different behavior between positions, indicating that the amplitude of activation, primarily of the EO and PFM, is independent of the position adopted for AHG. To date, no studies have been identified in the literature (MEDLINE/Pubmed, 1966–2016; LILACS,

Table 1
Median in microvolts (μV) and inter-quartile values (IQ), percentiles 25 (IQ25) and 75 (IQ75), of the maximum voluntary activity (MVC) during standardized tests, the maximum level of electromyography activity during the performance of abdominal hypopressive gymnastic (MVC AHG), and normalized electromyographic activity (%EMG) of the rectus abdominis (RA), external oblique (EO), transversus abdominis/internal oblique (TrA/IO) and the pelvic floor muscles (PFM) during the performance of hypopressive abdominal gymnastic in the supine, quadruped and orthostatic postures.

	SUPINE						QUADRUPED						ORTHOSTATIC					
	MVC		MVC AHG		%EMG		MVC AHG		%EMG		MVC AHG		%EMG			
	Median	IQ25 -IQ75	Median	IQ25 - IQ75	Median	IQ25 - IQ75	Median	IQ25 - IQ75	Median	IQ25 - IQ75	Median	IQ25 - IQ75	Median	IQ25 - IQ75	Median	IQ25 - IQ75
RA	112,60	83,25–147,15	2,57	1,07–3,67	4,44	2,50–7,68	3,62	1,46–7,18	7,42	4,45–12,71	3,68	1,64–6,51	6,63	3,53–9,17
EO	58,10	43,87–96,30	11,51	5,62–19,13	24,23	9,90–44,26	7,45	1,67–12,13	24,14	12,94–32,71	8,62	3,70–19,36	23,13	14,34–33,64
TrA/IO	39,60	25,17–73,30	13,68	7,20–24,41	51,83	28,40–81,44	13,10	2,55–25,67	59,96	24,39–83,16	15,77	1,75–35,25	77,04	25,83–106,37
PFM	40,25	30,42–61,07	7,57	4,52–12,77	43,69	35,52–65,37	10,30	1,91 - 1744	55,62	40,11–70,08	8,00	3,54–11,92	53,65	39,35–68,04

Table 2
Z-test values for Wilcoxon's post hoc test for difference between the values of normalized electromyographic activity (%EMG) between muscles rectus abdominis (RA), external oblique (EO), transversus abdominis/internal oblique (TrA/IO) and pelvic floor (MAP) and in each position analyzed.

%EMG				*z*	*P*
SUPINE	RA	vs.	EO	−4469	0,001**
	RA	vs.	TrA/IO	−4541	0,001**
	RA	vs.	PFM	−4541	0,001**
	EO	vs.	TrA/IO	−3671	0,001**
	EO	vs.	PFM	−2787	0,005**
	TrA/IO	vs.	PFM	−1162	0,245
QUADRUPED	RA	vs.	EO	−4356	0,001**
	RA	vs.	TrA/IO	−4457	0,001**
	RA	vs.	PFM	−4432	0,001**
	EO	vs.	TrA/IO	−2433	0,015*
	EO	vs.	PFM	−3298	0,001**
	TrA/IO	vs.	PFM	−0,465	0642
ORTHOSTATIC	RA*	vs.	EO	−4167	0,001**
	RA	vs.	TrA/IO	−4623	0,001**
	RA	vs.	PFM	−4600	0,001**
	EO	vs.	TrA/IO	−3486	0,001**
	EO	vs.	PFM	−3281	0,001**
	TrA/IO	vs.	PFM	−1944	0,052

$p < 0,05$.
$^{**}p < 0,01$.

1982–2016) that assess the effect of body position on the amplitude of electromyographic activity of the AM and PFM during abdominal hypopressive gymnastics. Madill and McLean (2008) suggest that vertical positions contribute to the increase in PFM activity in order to compensate for the rise in intra-abdominal pressure in these positions (Stéphanie J. Madill and McLean, 2008). However, when comparing between the amplitude of the AM and PFM in the supine decubitus, sitting and orthostatic positions, during the recording of maximal voluntary contraction of the PFM, the authors found no difference between muscle activation amplitude in the positions analyzed, indicating that gravity has no effect on the activation capacity of the PFM.

To date, only one study has used surface electromyography to assess AM and PFM activation (Stüpp et al., 2011). The authors investigated a group of nulliparous physical therapists with respect to the capacity of abdominal hypopressive gymnastics in the supine position to promote PFM activation above resting muscle tonus, in addition to determining if AHG could be comparable to training using voluntary contraction of the PFM. They observed that the increase in relation to basal tonus was significantly higher, but voluntary contraction of the PFM caused a significantly higher increase than AHG. Even though the study presented electromyographic readings of the TrA and PFM in the supine position, it was impossible to compare it with the present investigation given that no normalization was applied.

It is important to point out a number of limitations. The characteristics of the recordings obtained precluded an assessment of response time between the muscles evaluated. Thus, it was impossible to determine the temporal sequence of recruitment of these muscles during AHG. Thus, the use of surface electrodes precluded distinguishing between the TrA and IO muscles at the site used to assess them. Given the absence of standardized exercises to obtain greater muscle activity in the quadruped and orthostatic positions, the use of a reference in the supine position to normalize electromyographic activity in all the positions assessed may not reliably represent the relative percentage of electromyographic activation of these muscles during AHG, in the quadruped and orthostatic positions.

Furthermore, although it was not our goal to measure the respiratory muscles electromyographic activity, we also recognize

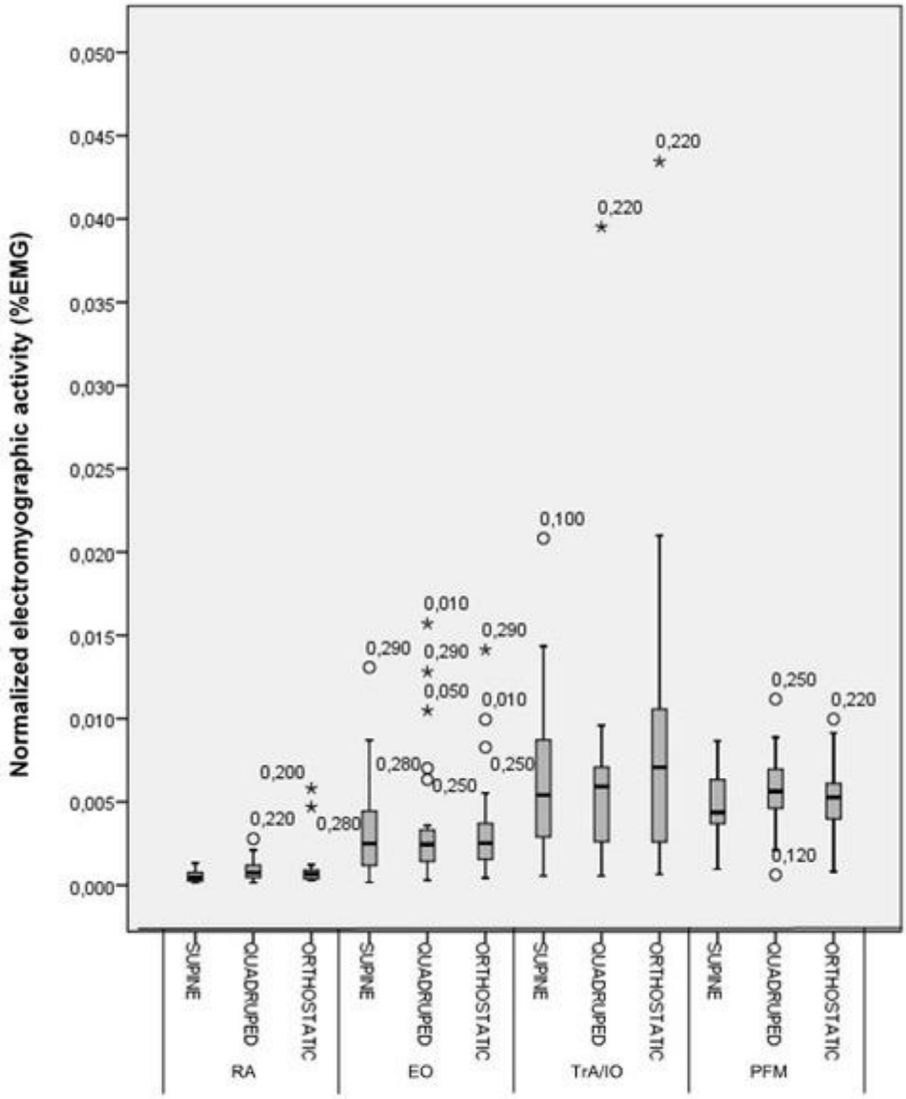

Fig. 5. Normalized electromyographic activity (EMG%) during the performance of abdominal hypopressive gymnastic in the decubitus supine, quadruped and orthostatic postures. The box plot indicate the median in microvolts (μV) and inter-quartile values of EMG% of the rectus abdominis (RA), external oblique (EO), transversus abdominis/internal oblique (TrA/IO) and pelvic floor muscles (PFM).

that the diaphragm and others important respiratory muscles, such as intercostal muscles, scalene muscles and sternocleidomastoid may play role during AHG.

The highest level of activation and synergism observed in muscles TrA/IO and PFM while performing the abdominal hipopressive gymnastic suggests that this technique may be used in situations requiring coactivation of these muscles, as in dysfunctions of the pelvic floor and lumbopelvic postural disorders.

It is important to emphasizing there are some limitations and contraindications of this technique for be prescribed or recommended as a treatment protocol. For example, in case of pregnancy, hypertension, neuromuscular, cardiac and respiratory diseases (Caufriez et al., 2011, 2007, 2006). Others potential concerns not described in the literature about others clinical repercussions of the AHG should also be pointed once this technique involves the raises and fluctuations in the intrathoracic and intra-thecal pressure.

5. Conclusion

This study demonstrated that abdominal hypopressive gymnastics is capable of activating the abdominal muscles, primarily the TrA/IO and PFM. Similar activation patterns were identified in the supine, quadruped and orthostatic positions for PFM and EO. Furthermore, TrA/IO muscle exhibits significant differences across positions with highest activity in standing positions.

Future studies should assess the influence of pelvic position and upper limbs, the temporal sequence of recruitment of AM and PFM, as well as whether there is an alteration in activation level exhibited by these muscles in a population with pelvic floor dysfunction. Furthermore, the possible effects of abdominal hypopressive gymnastics on respiratory muscles and lung function should also be explored, since AHG results primarily from accessory inspiratory muscle activation associated to expiratory apnea.

Declaration of interest

Conflicts of interest

None.

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References

Auchincloss, C., McLean, L., 2012. Does the presence of a vaginal probe alter pelvic floor muscle activation in young, continent women? J. Electromyogr. Kinesiol 22, 1003–1009. http://dx.doi.org/10.1016/j.jelekin.2012.06.006.

Bernardes, B.T., Resende, A.P.M., Stüpp, L., Oliveira, E., Castro, R.A., Bella, Z.I.K.J., di, Girão, M.J.B.C., Sartori, M.G.F., 2012. Efficacy of pelvic floor muscle training and hypopressive exercises for treating pelvic organ prolapse in women: randomized controlled trial. Sao Paulo Med. J. 130, 5–9.

Caufriez, M., 1995. Techniques abdominales hypopressives et rééducation uro-gynécologique. Kinésithérapie Sci. 1, 53–55.

Caufriez, M., 1988. Thérapies manueles et instrumentales en uro-gynécologie, bases théorics et test. Brus. Maïte 1.

Caufriez, M., 1989. Thérapies manueles et instrumentales em uro-gynécologie, pratiques spécifiques. Brus. Maïte 2.

Caufriez, M., Fernández-Domínguez, J.C., Brynhildsvoll, N., 2011. Estudio preliminar sobre la acción de la gimnasia hipopresiva en el tratamiento de la escoliosis idiopática. Enfermería Clínica 21, 354–358. http://dx.doi.org/10.1016/j.enfcli.2011.06.003.

Caufriez, M., Fernández, J.C., Fanzel, R., Snoeck, T., 2006. Efectos de un programa de entrenamiento estructurado de Gimnasia Abdominal Hipopresiva sobre la estática vertebral cervical y dorsolumbar. Fisioterapia 28, 205–216. http://dx.doi.org/10.1016/S0211-5638(06)74048-2.

Caufriez, M., Fernández, J.C., Guignel, G., Heimann, A., 2007. Comparación de las variaciones de presión abdominal en medio acuático y aéreo durante la realización de cuatro ejercicios abdominales hipopresivos. Rev. Iberoam. Fisioter. Kinesiol 10, 12–23. http://dx.doi.org/10.1016/S1138-6045(07)73661-4.

Chanthapetch, P., Kanlayanaphotporn, R., Gaogasigam, C., Chiradejnant, A., 2009. Abdominal muscle activity during abdominal hollowing in four starting positions. Man. Ther. 14, 642–646. http://dx.doi.org/10.1016/j.math.2008.12.009.

Hodges, P., Cresswell, A., Thorstensson, A., 1999. Preparatory trunk motion accompanies rapid upper limb movement. Exp. Brain Res. 124, 69–79.

Junginger, B., Baessler, K., Sapsford, R., Hodges, P.W., 2010. Effect of abdominal and pelvic floor tasks on muscle activity, abdominal pressure and bladder neck. Int. Urogynecol. J. 21, 69–77. http://dx.doi.org/10.1007/s00192-009-0981-z.

Keshwani, N., McLean, L., 2013. A differential suction electrode for recording electromyographic activity from the pelvic floor muscles: crosstalk evaluation. J. Electromyogr. Kinesiol 23, 311–318. http://dx.doi.org/10.1016/j.jelekin.2012.10.016.

Madill, S.J., McLean, L., 2008. Quantification of abdominal and pelvic floor muscle synergies in response to voluntary pelvic floor muscle contractions. J. Electromyogr. Kinesiol 18, 955–964. http://dx.doi.org/10.1016/j.jelekin.2007.05.001.

Marshall, P., Murphy, B., 2003. The validity and reliability of surface EMG to assess the neuromuscular response of the abdominal muscles to rapid limb movement. J. Electromyogr. Kinesiol 13, 477–489. http://dx.doi.org/10.1016/S1050-6411(03)00027-0.

Matsudo, S., Araújo, T., Matsudo, V., Andrade, D., Andrade, E., Oliveira, L.C., Braggion, G., 2001. Questionário internacional De atividade física (ipaq): estupo De validade e reprodutibilidade No Brasil. Rev. Bras. Atividade Física Saúde 6, 5–18.

Neumann, P., Gill, V., 2002. Pelvic floor and abdominal muscle interaction: EMG activity and intra-abdominal pressure. Int. Urogynecol. J. 13, 125–132. http://dx.doi.org/10.1007/s001920200027.

Ng, J.K., Kippers, V., Richardson, C.A., 1998. Muscle fibre orientation of abdominal muscles and suggested surface EMG electrode positions. Electromyogr. Clin. Neurophysiol. 38, 51–58.

Resende, A.P.M., Stüpp, L., Bernardes, B.T., Oliveira, E., Castro, R.A., Girão, M.J.B.C., Sartori, M.G.F., 2012. Can hypopressive exercises provide additional benefits to pelvic floor muscle training in women with pelvic organ prolapse? Neurourol. Urodyn. 31, 121–125. http://dx.doi.org/10.1002/nau.21149.

Rial, T., Villanueva, C., 2012. La gimnasia hipopresiva en un contexto de actividad físico-saludable y preventiva. Rev. Transm. del Conoc. Educ. Salud 4, 215–230.

Sapsford, R.R., Hodges, P.W., 2001. Contraction of the pelvic floor muscles during abdominal maneuvers. Arch. Phys. Med. Rehabil. 82, 1081–1088. http://dx.doi.org/10.1053/apmr.2001.24297.

Sapsford, R.R., Hodges, P.W., Richardson, C.A., Cooper, D.H., Markwell, S.J., Jull, G.A., 2001. Co-activation of the abdominal and pelvic floor muscles during voluntary exercises. Neurourol. Urodyn. 20, 31–42. http://dx.doi.org/10.1002/1520-6777(2001)20, 1<31::AID-NAU5>3.0.CO;2-P.

Stüpp, L., Resende, A.P.M., Petricelli, C.D., Nakamura, M.U., Alexandre, S.M., Zanetti, M.R.D., 2011. Pelvic floor muscle and transversus abdominis activation in abdominal hypopressive technique through surface electromyography. Neurourol. Urodyn. 30, 1518–1521. http://dx.doi.org/10.1002/nau.21151.

Taillon-Hobson, A., Aaron, S., McLean, L., Bilodeau, M., 2011. Voluntary and automatic recruitment of superficial and deep abdominal muscles in adults with and without cystic fibrosis. J. Electromyogr. Kinesiol 21, 411–417. http://dx.doi.org/10.1016/j.jelekin.2011.02.006.

Torres, G., Salido, A., 2009. Programas de ejercicio en lumbalgia mecanopostural. Rev. Mex. Física Rehabil 21, 11–19.

Vera-Garcia, F.J., Moreside, J.M., McGill, S.M., 2011. Abdominal muscle activation changes if the purpose is to control pelvis motion or thorax motion. J. Electromyogr. Kinesiol 21, 893–903. http://dx.doi.org/10.1016/j.jelekin.2011.08.003.